MIL-P-60242(MU)
AMENDMENT 3
25 June 1968
SUPERSEDING
AMENDMENT 2
20 February 1968

MILITARY SPECIFICATION

PERISCOPE, TANK: XM51

This amendment forms a part of Military Specification MIL-P-60242(MU), dated 28 February 1966.

Page 1: Delete "NOT TO BE RELEASED FOR PUBLICATION OR PUBLIC DISPLAY." (Specifications reproduced for replenishment after February 1968 do not carry this notice.)

* Paragraph 3.10.3(b): Delete and substitute:

"3.10.3.(b) With the head positioned as in 3.10.1 and the line of sight coincident with the horizontal plane; the differential angular motion between the line of sight through the head mirror and the relative rotation of the synchro drive shaft shall provide 355.21 ± 0.05 mils rotation of the synchro shaft for 355.00 mils movement of the line of sight above the horizontal plane."

Paragraph 3.11.3.1.1, Table II, under "NOTE": After "2870°" add "±100°K" and make this same addition wherever "2870°K" is used in this specification.

Paragraph 3.11.3.2.1b: Delete in its entirety, including Table IV.

Paragraph 3.11.3.4.1.1b: Delete in its entirety.

Paragraph 3.11.3.6: Delete "$0.02G_T$" and substitute "$0.013G_T$"; Delete the "NOTE" in its entirety and substitute: "NOTE" G_T is the minimum luminance gain value specified in the image intensifier tube purchase description or proposed military specification applicable to the tube when manufactured."

Paragraph 3.11.3.8: Delete "2.0 percent" and substitute "20.0 per cent".

Paragraph 3.11.4.2, 2nd sentence: Delete and substitute: "The force required to disengage the passive daylight lever detent shall not exceed 12 pounds between -40°F and +150°F."

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Paragraph 3.11.4.3.1: Delete and substitute:

"3.11 .4.3.1 Rheostat control. - The running torque required to rotate the rheostat control knob through temperatures ranging from -40° to +150°F shall not exceed 5 inch pounds."

Paragraph 3.12.2.6: Delete "0.075 mil" and substitute "0.1 mil".

Paragraph 3.13.1: Add the following:

On line "d", after "3.11.3.1" add "& 3.11.3.6"
On line "f", after "3.11.3.4" add "3.11.3.5"
Under "Requirement" and "Paragraph" add the following:

"m" Luminance gain 3.11.3.7"

Paragraph 4.3.2, Tables VIII, IX, X & XI: Delete "AQL 0.657. defective" and substitute "AQL 1.0%. defective.

Paragraph 4.3.2, Table VIII: Delete Characteristic "108".

Paragraph 4.3.2, Table IX: Delete Characteristics "114, 115, 117, 121; 124, 125 and 126".

Paragraph 4.3.2, Table XI: Delete Characteristics "169" and "171".

Add the following paragraphs:

"4.3.2.1 Body assembly. - All body assemblies produced (100% inspection) as repair parts, including those subjected to Tables IX and XIV shall be inspected for the following characteristics:

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
a.	Contrast transfer, objective system, on-axis	3.11.3.1.1	4.6.5.1.1
b.	Contrast transfer, eyepiece and erecting system, on-axis	3.11.3.1.2	4.6.5.1.3
c.	Objective T-number, on-axis	3.11.3.4.1.1a	4.6.5.1.11
d.	Contrast transfer, on-axis	3.11.3.6	4.6.5.1.18
e.	Luminance gain	3.11.3.7	4.6.5.1.20

4.3.2.2 Periscope, Tank, XM51. - All periscopes produced (100% inspection) including those subjected to Tables XI and XVI, shall be inspected for the following characteristics:

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	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
a.	Contrast transfer, objective system, eyepiece and erecting system, on-axis	3.11.1d (3.11.3.1)	4.6.7.1.1
b.	Contrast transfer, on-axis	3.13.1e (3.11.3.6)	4.6.5.1.18
c.	Resolution objective T-number on-axis	3.13.1g (3.11.3.4.1.1a)	4.6.5.1.1
d.	Luminance gain	3.13.in (3.11.3.7)	4.6.5.1.20"

Paragraphs 4.4.2.1, 4.4.2.2, 4.4.2.3 and 4.4.2.4, line 1: delete "Three" and substitute "TWO".

Paragraph 4.4.2.2, after heading, add: "When provided as a repair part assembly (separate from periscopes), two ----"; At end of 2nd sentence, add: "and 4.3.2.1".

Paragraph 4.4.2.2, Table XIV: Delete Characteristics "312" and "313"; Characteristics "311" and "314" and substitute the following:

"311.	Resolution, objective system (+60° to +90°F)	3.11.3.4.1a & b	4.6.5.1.9 & 4.6.5.1.10
314	Resolution, passive	3.11.3.5	4.6.5.1.16 4.6.5.1.17"

Paragraph 4.4.2.4, 2nd sentence: Add "and 4.3.2.2" to end of sentence.

Paragraph 4.4.2.4, Table XVI: Delete Characteristic "324"; add new Characteristic as follows:

"324.	Resolution, passive system (+60° to +90°F)	3.13.1f (3.11.3.5)	4.6.7.1.4, 4.6.5.1.16, 4.6.5.1.17"

* Paragraph 4.6.4.1.3(b), line 14: Delete "20 degrees" and substitute "355.00 mils".

* Paragraph 4.6.4.1.3(b), lines 15 and 16: Delete "...not deviate beyond the tolerance . .." and substitute ... be as . ..".

Paragraph 4.6.4.1.7: Delete in its entirety.

Paragraph 4.6.5.1.5: Delete in its entirety.

Paragraph 4.6.5.1.6, line 2: Delete "4.6.5.1.5" and substitute "4.6.5.1.4".

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Paragraph 4.6.5.1.12: Delete in its entirety.

Paragraph 4.6.5.1.15, "NOTE": Delete the note in its entirety.

Paragraph 4.6.5.1.16: Add the following "caution" paragraph.

"Caution: The image tube must be erased (completely discharged), and the observer dark adapted for not less than 15 minutes before performing the low contrast resolution test."

Paragraph 4.6.5.1.21, line 4: Delete "10^{-6}" and substitute "10^{-4}"; line 8, delete "10^{-3}" and substitute "10^{-2}".

Paragraph 4.6.7.1.2, line 2: Delete "4.6.5.1.5".

Paragraph 4.6.7.1.3, line 2: Delete "4.6.5.1.12".

Paragraph 6.2: Add "h" as follows:

"h. Requirements for reinspection of image intensifier tubes by the contractor, when provided as Government Furnished Items (GFI), prior to insertion into periscopes for final acceptance inspection."

* The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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